Initial decision record

Recommendation to trustees · Decision date 31 Dec 2022 · Prepared for 22 Oct deadline

DECISION / Fund the two fixed commitments with case bonds, then invest the residual in USMV.

Payment security comes first; the USD 8m December-2022 purchasing-power target for 2030 is an objective, not a promise. Review risk on the growth sleeve and payment coverage separately.

01 CAPITAL AND OBLIGATIONS

Initial use	Case instrument / purpose	USD	% of fund
2024 commitment	2,400 Beta zeros; USD 2.4m face	$2,214,672.00	18.456%
2027 commitment	3,478.26087 Delta; final USD 3.6m	$3,429,808.70	28.582%
Growth sleeve	USMV, after entry friction	$6,346,350.71	52.886%
Execution costs	5bp bonds + 10bp equity	$9,168.59	0.076%
Total initial cash	Fully allocated	**$12,000,000.00**	100.000%

Coverage: Beta pays USD 2.4m on 31 Dec 2024; Delta's last coupon plus principal pays USD 3.6m on 31 Dec 2027. Its four earlier coupons total USD 486,956.52 and stay in cash at 0% base reinvestment. No equity sale or assumed bond resale price funds either obligation.

02 CANDIDATE SET, HORIZON AND COMPARATOR

Eight US stocks / seven sectors: AAPL, MSFT (technology); JPM (financials); JNJ (health care); PG (staples); XOM (energy); CAT (industrials); NEE (utilities). **Two distinct US equity ETFs:** USMV (US minimum-volatility factor) and VTI (broad US market). They are candidates, not required holdings; pre-2023 SEC eligibility references are in the data record.

Estimation: 96 common monthly adjusted-price returns, Jan 2015-Dec 2022; Dec 2014 closes supply the first return baseline. Test 60 months as robustness later. **Investable growth comparator:** 100% VTI, frozen on the same 2022 decision date and evaluated separately from the case bonds.

03 GROWTH PORTFOLIO: SELECTION AND REJECTED ALTERNATIVE

Selected weights inside the growth sleeve: USMV 100% (USD 6,346,350.71); VTI 0%; eight direct-stock candidates 0%. VTI is the frozen, investable comparator, not a second holding. The stock list is a research universe: the initial allocation does not depend on repeating any individual stock's historical Sharpe ratio or estimated alpha.

2022-input comparison	USMV · recommended	VTI · credible alternative
CAPM annual return scenario	7.14%	8.45%
2015-22 annualized volatility	12.84%	16.31%
Utility at A = 4	3.84%	3.13%

Decision logic: USMV spreads the sleeve across US stocks with a minimum-volatility mandate and has lower 2015-22 modeled volatility than broad-market VTI. Utility = expected annual return - ½A × annual variance. A = 4 is **provisional**; VTI ranks higher at A = 2, USMV at A = 4/6/8, with a two-ETF crossover near A = 2.58. VTI is a credible rejected growth allocation because of its higher forecast return. USMV can still lose more than 15%; its arithmetic forecast is not a guarantee of the required eight-year CAGR or a global portfolio optimum.

Policy, evidence and controls

Only the 2022 information set supports the initial decision · All thresholds below are proposals

04 MATERIAL ASSUMPTIONS AND LIMITS

Topic	Proposed convention / decision use
Real goal	CPI planning 2.5%: 2030 nominal USD 9.747m; USMV sleeve needs 4.84% nominal CAGR with coupon cash at 0%. Education-cost 3.0% stress: USD 10.134m / 5.37% hurdle. Do not blend CPI and HEPI.
Return & risk	CAPM scenario: annual RF 4%, market premium 4.5%, forecast alpha 0; monthly French RF is used for historical excess returns. FF3 explains historical exposures, not a factor-premium forecast. Compare A = 2, 4, 6, 8.
Instruments / costs	Case permits cash, its five bonds, US common stocks and US equity ETFs; no leverage, shorts or derivatives. Fractional bonds allowed. Base friction: 5bp bonds, 10bp equity; 0bp bonds / 5bp ETFs is a sensitivity, not the base.

No additional stock-position cap has been approved. The two mandatory nominal payments are covered by contractual cash flows under the case's default-free assumption; the 2030 real goal, modeled returns and a maximum loss cannot be guaranteed.

05 MONITORING, DOWNSIDE AND UPSIDE

Primary signal: **unitized growth-sleeve total-return drawdown** from its running high; adjust external transfers, include sleeve cash and do not reset the high-water mark after switching assets. Track whole-fund value, dated bond-payment coverage and 2030 real-goal feasibility as separate measures. Monthly historical data support month-end signals only.

Signal	Required governance response · proposed, not a price stop
10% drawdown	Within 2 business days, validate data, payment coverage and stress. No automatic transaction.
15% drawdown	Committee meets within 5 business days; record a hold, authorized rebalance or prospective policy recommendation.
20% drawdown	Notify trustees same/next business day; pause discretionary risk increases. Do not force an equity sale.
Payment gap · any level	Escalate immediately with a funded corrective plan. Crisis exceptions cannot weaken due-payment coverage.

Upside review: annually recheck liabilities, the CPI goal and a -30% growth-sleeve shock. If the goal remains fundable after that stress, trustees may prospectively protect only the minimum required amount in an eligible, actually priced asset matched to 2030; keep surplus invested. No fixed-gain sale or automatic spending increase. This requires a documented forward decision, not a rewrite of the frozen backtest.

06 FROZEN EVALUATION RULE AND DECISION CONTROL

Freeze USMV and VTI at 31 Dec 2022. For Jan 2023-Dec 2025, compare the growth sleeve and benchmark using a declared annual post-December-return reset to inception weights, **without re-estimating** parameters; a one-ETF sleeve normally has no internal trade. Report costs and unfavorable evidence. Review payment coverage each year; changed mandates, assumptions or allocation need a dated before/after analysis and prospective effective date. Do not reset payment bonds to an initial percentage after they pay.

07 REPRODUCE AND INSPECT THE DATED DATA RECORD

1. Read data/DATA_RECORD.md and data/raw/retrieval_manifest.jsonl for source URLs, UTC timestamps, request settings, package versions and SHA-256. **2.** Install requirements-analysis.txt, then run python analysis/run_initial.py from the archive root, offline. **3.** Inspect initial_decision_rebuild.json (budget/goal), growth_portfolio_weights.csv (holdings), candidate_statistics.csv (CAPM) and risk_preference.csv (A sensitivity), all in outputs/. Online retrieval: analysis/retrieve_data.py --phase train.

Source caveat: saved Yahoo/French downloads were retrieved in Oct 2026 and may contain revised history; this is a retrospective case, not a genuinely unseen 2022 trade. Ten Yahoo raw CSVs + one French ZIP passed saved-byte hash and 96-month checks. SEC filing checks are dated links, not locally saved SEC bytes (direct downloads returned HTTP 403). The 2023-25 evaluation is outside this initial record.

Scope of this two-page record: detailed bond duration/curve shocks, sample-versus-single-index optimization, multifactor alpha tests and 2023-25 outcomes belong in the final committee memo; none is claimed as completed by this initial record.

Practice reference: University of California, General Endowment Pool Investment Policy Statement (2012), pp. 3-4, 7-8: mission, fiduciary oversight, benchmarks and reporting. Kappa's numerical rules are case-specific proposals.
https://www.ucop.edu/investment-office/_files/invpol/GEP_IPS.pdf